

ECE 350
Real-time
Operating
Systems



Lecture 3: Multithreaded Kernels

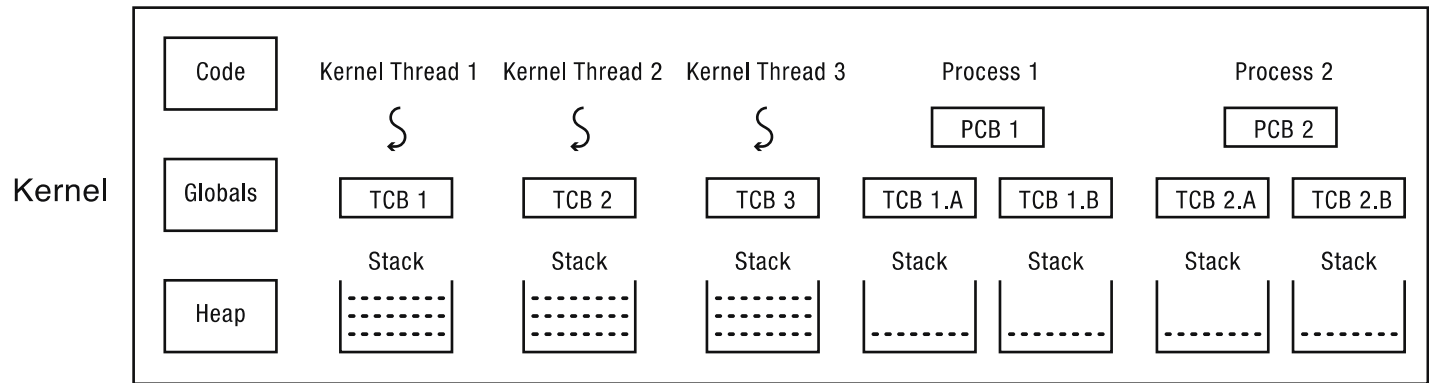
Prof. Seyed Majid Zahedi

<https://ece.uwaterloo.ca/~smzahedi>

Outline

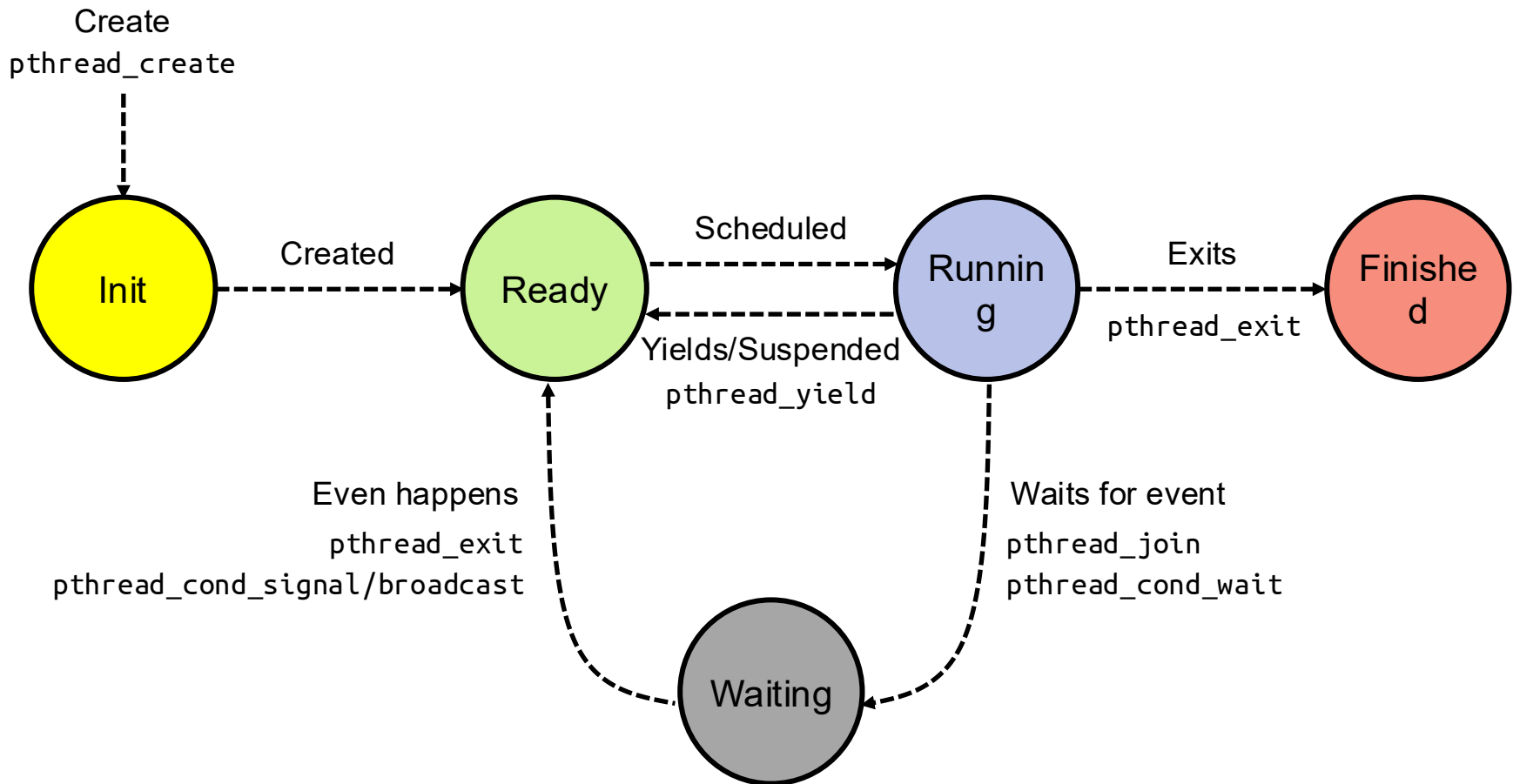
- Thread implementation
 - Create, yield, switch, etc.
- Kernel- vs. user-managed threads
- Implementation of synchronization objects
 - Mutex, semaphore, condition variable

Kernel-managed Multithreading



- System library allocates **user-space stack** for each user-level thread
- System library then uses system calls to create, join, yield, exit threads
- Kernel handles scheduling and context switching using **kernel-space stacks**

Recall: Thread Lifecycle



A process can go directly from ready or waiting to finished (example: main thread calls exit)

What Triggers a Context Switch?

- **Synchronous event:** thread invokes a system call or an exceptions happens

- E.g., yield, join, write, read, or segmentation fault, divide by zero, etc.

- This is called a voluntary context switch

```
void compute_PI() {  
    while(TRUE) {  
        compute_next_digit();  
        thread_yield();  
    }  
}
```

- **Asynchronous event:** interrupts happens
 - E.g., timer interrupt, new packet arrives, a DMA request finishes
 - This is called an involuntary context switch

System Call, Interrupt, and Exception Handlers

```
handler() {  
    // this runs in kernel mode  
    // SP points to a kernel stack  
    Push regs that might be used by handler on kernel stack  
  
    // (handle the event)  
  
    Handler_Exit  
        Pop regs that were pushed  
        Return  
}
```



Switch Between Threads

```
// We enter as oldTCB, but we return as newTCB
// Returns with newTCB's registers and stack
```

```
thread_switch(TCB *oldTCB, TCB *newTCB) {
    Push all regs onto kernel stack of oldTCB
    Set oldTCB->sp to stack pointer
    Set stack point to newTCB->sp
    Pop regs from kernel stack of newTCB
    Return
}
```

newTCB could be a thread that was context switched before and we are context switching back to it, or it could be a **newly created** thread

Where does this return to?

If newTCB is not newly created, then we return to **kernel code** that called thread_switch (return address is stored on newTCB's stack)

If newTCB is newly created, then it should have an **entry point** address on its stack

What is popped here?

If newTCB is not newly created, then we pop what we pushed **last time** we context switched it

If newTCB is newly created, then it should have **dummy data frame** on top of its stack

Threads Entry Point

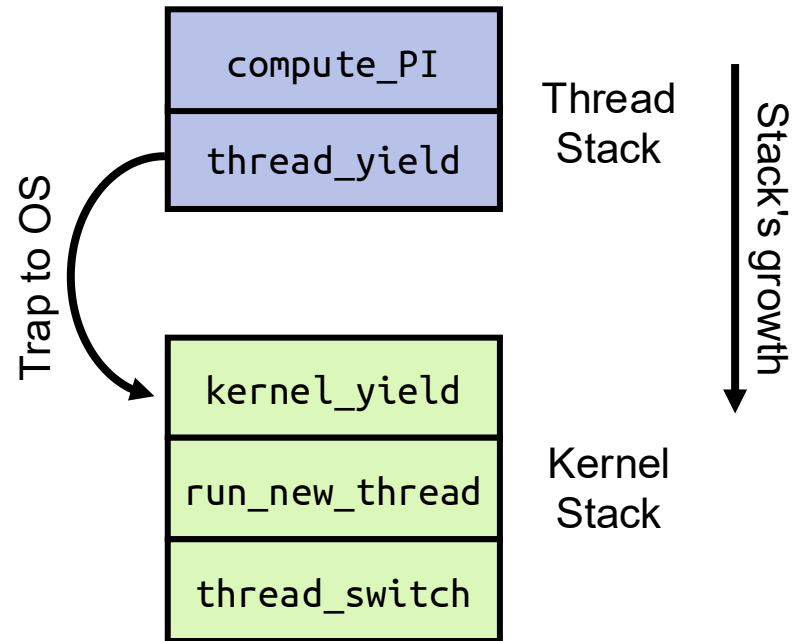
- For kernel threads, no mode switch is required
 - Could directly jump to function that thread will run
- For user threads, switch from kernel to user mode is required
 - Need one level of indirection
 - Could jump to a kernel code that then jumps to user code and changes mode atomically
 - E.g., could jump to `Handler_Exit`

Creating New User Threads

```
thread_create(void *(*func)(void*), void *args) {  
    // Allocate TCB  
    TCB *tcb = new TCB()  
    // Allocate kernel stack (note that stack grows downwards)  
    tcb->sp = new Stack(stack_size) + stack_size;  
    // Set up kernel stack  
    // (1) Push func and args  
    * (--tcb->sp) = args;  
    * (--tcb->sp) = func;  
    // (2) push data for Handle_Exit  
    push_dummy_handler_frame(&tcb->sp);  
    * (--tcb->sp) = Handler_Exit  
    // (3) Push dummy data for thread_switch  
    push_dummy_switch_frame(&tcb->sp);  
    // Set state of thread to read  
    tcb->state = READY;  
    // Put tcb on ready list  
    readyList.add(tcb);  
}
```

Stack for Yielding Thread

```
void run_new_thread() {  
    // Prevent interrupt from stopping us  
    // in the middle of switch  
    disable_interrupts();  
    // Choose another TCB from ready list  
    chosenTCB = scheduler.getNextTCB();  
    if (chosenTCB != runningTCB) {  
        // Move running thread onto ready list  
        runningTCB->state = READY;  
        ready_list.add(runningTCB);  
        // Switch to the new thread  
        thread_switch(runningTCB, chosenTCB);  
  
        // We're running again!  
        runningTCB->state = RUNNING;  
        // Do any cleanup  
        do_cleanup_housekeeping();  
    }  
    // Enable interrupts again  
    enable_interrupts();  
}
```



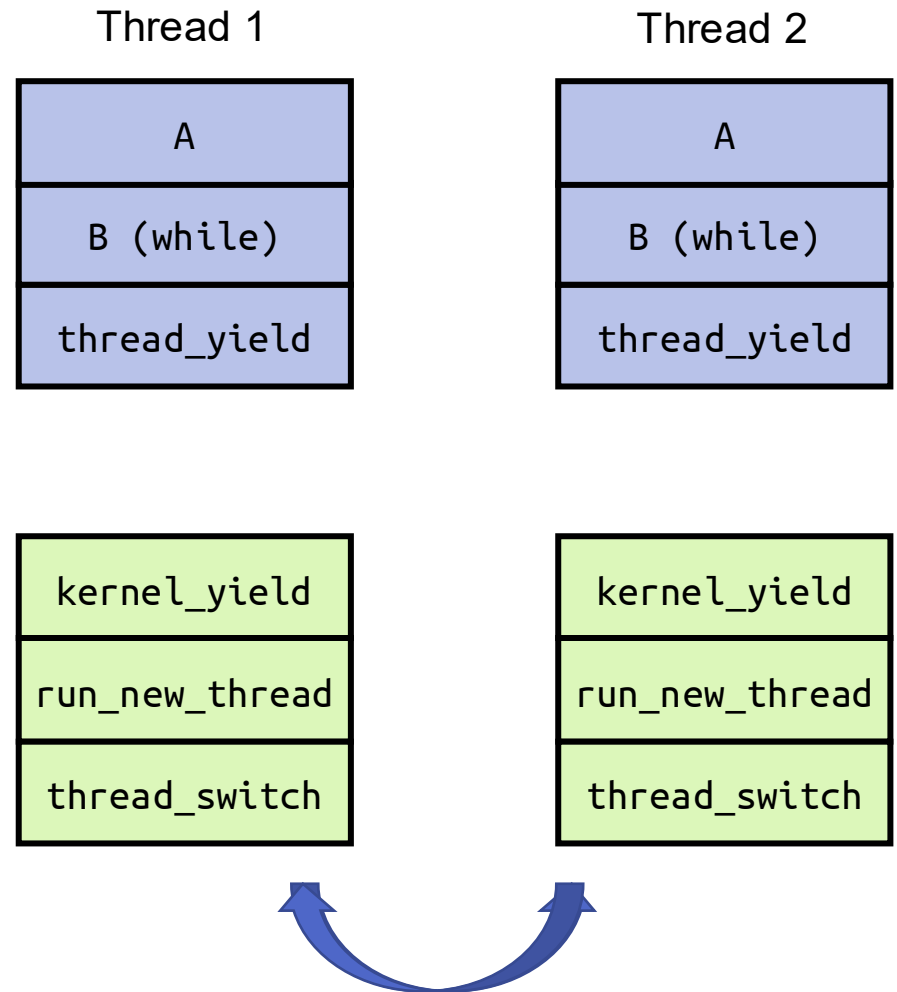
Start from here whenever another thread switches back to this thread

How Do Stacks Look Like?

- Two threads run following code

```
A() {  
    B();  
}
```

```
B() {  
    while(TRUE) {  
        thread_yield();  
    }  
}
```

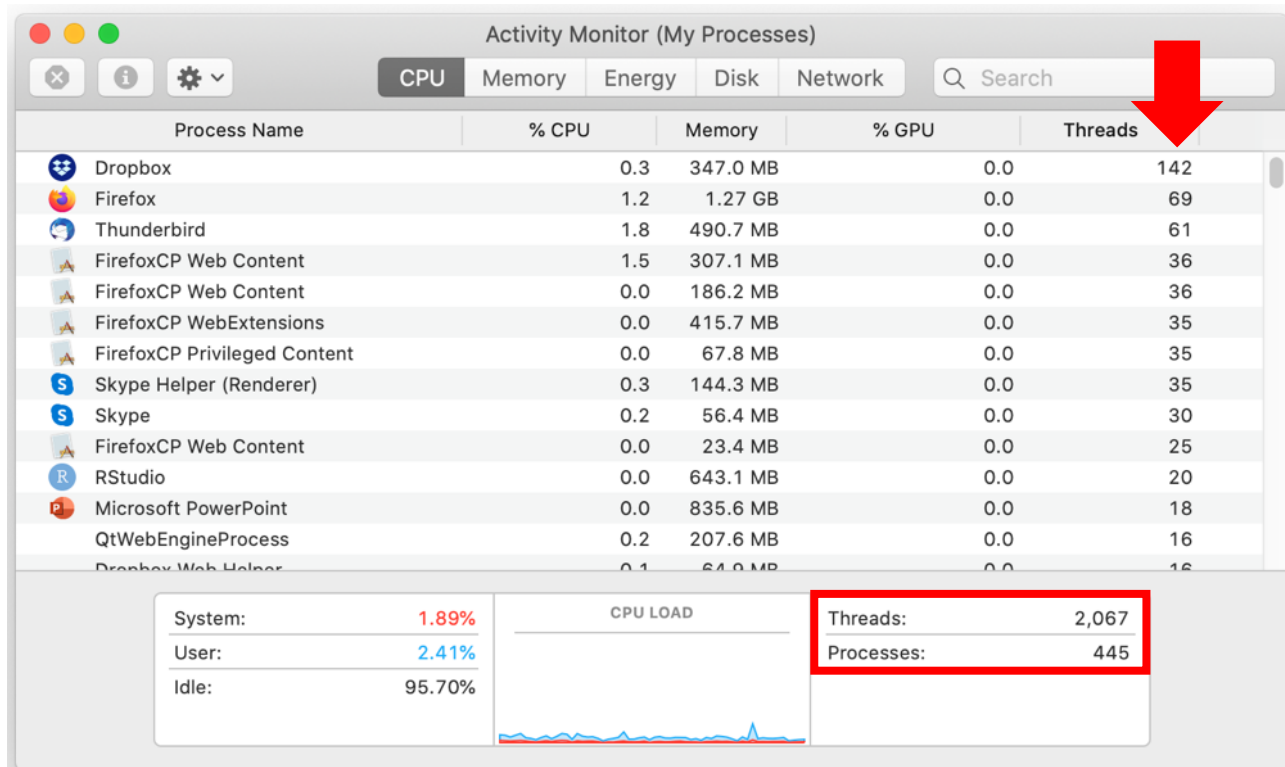


Outline

- Thread implementation
 - Create, yield, switch, etc.
- **Kernel- vs. user-managed threads**
- Implementation of synchronization objects
 - Mutex, semaphore, condition variable

Some Numbers

- Many process are **multi-threaded**, so thread context switches may be either **within-process** or **across-processes**

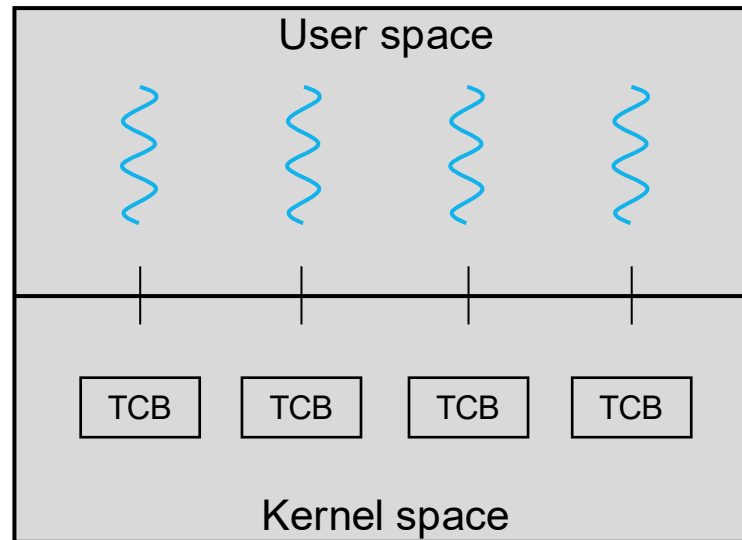


Some Numbers (cont.)

- Frequency of performing context switches is ~10-100ms
- Context switch time in Linux is ~3-4 us (Intel i7 & Xeon E5)
 - Thread switching faster than process switching (~100 ns)
- Switching across cores is ~2x more expensive than within-core
- Context switch time increases sharply with size of working set*
 - Can increase ~100x or more
- Moral: overhead of context switching depends mostly on cache limits and process or thread's hunger for memory

* Working set is subset of memory used by process in time window

Kernel- vs. User-managed Threads



- We have been talking about kernel-managed threads
- Each user thread maps to one TCB (1:1 mapping)
- Every thread can run or block independently
- This approach is relatively expensive
 - Need to make crossing into kernel mode to schedule

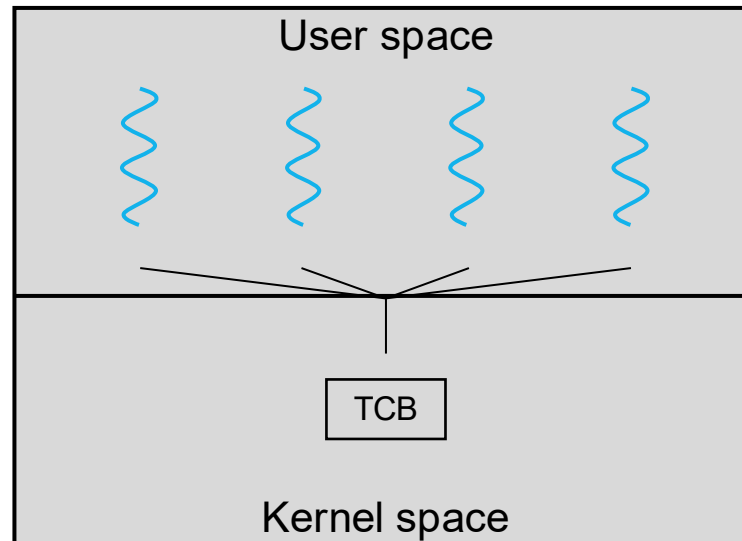
Kernel-Managed Threads

- **Scheduling**
 - Kernel scheduler decides when they run on the CPU
- **Context switching**
 - Kernel saves/restores CPU state during switches
- **Blocking**
 - If one thread blocks (e.g., on I/O), kernel can run another
- **Parallelism**
 - Can run truly in parallel on multiple CPU cores

Kernel-Managed Threads (cont.)

- **Pros:**
 - True parallelism on multi-core systems
 - Blocking one thread doesn't block others in same process
 - Kernel can preempt them fairly
- **Cons:**
 - Thread creation/switching is expensive (requires system calls, mode switches)
 - Memory overhead per thread (kernel stack, kernel data structures)

User-managed Threads



- Alternative is for user-level library to do all thread management tasks
- User process creates threads, maintains their state, and schedules them
- Kernel is not aware of existence of multiple threads
- Kernel only allocates single TCB to user process (N:1 mapping)

User-Managed Threads

- **Scheduling**
 - Done by a user-space scheduler (e.g., a runtime)
- **Context switching**
 - Cheap, no kernel mode switch needed
- **Blocking**
 - If one user-managed "thread" blocks on a real syscall, the entire kernel-managed thread may block
- **Parallelism**
 - Limited; often run concurrently but not truly in parallel (unless mapped to multiple kernel threads)

User-Managed Threads (cont.)

- **Pros:**

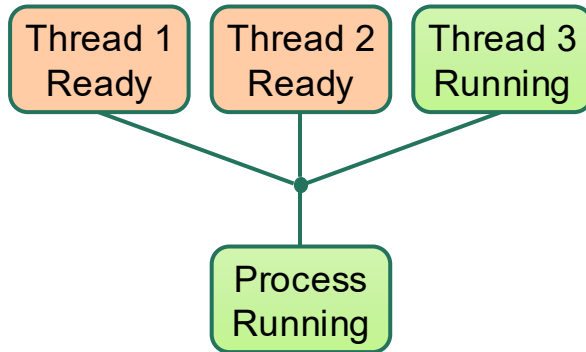
- Very fast to create and switch between (no syscall)
- Can have thousands or millions of them cheaply
- Programmer controls scheduling logic

- **Cons:**

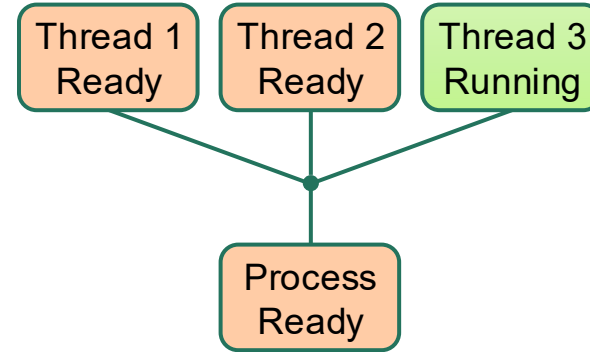
- A blocking syscall can stall the whole process
- Doesn't automatically leverage multiple CPU cores
- Requires cooperative yielding (or a smart runtime to work around limitations)

User-managed Threads: Thread vs. Process State

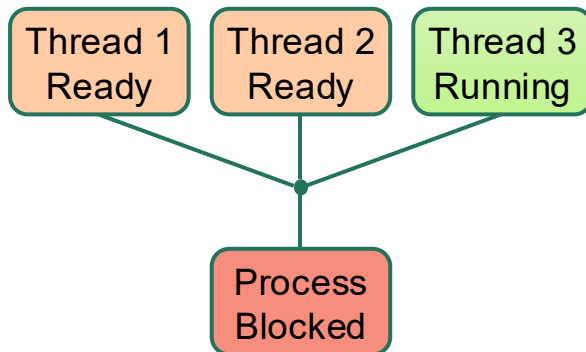
Thread 3 is running on CPU



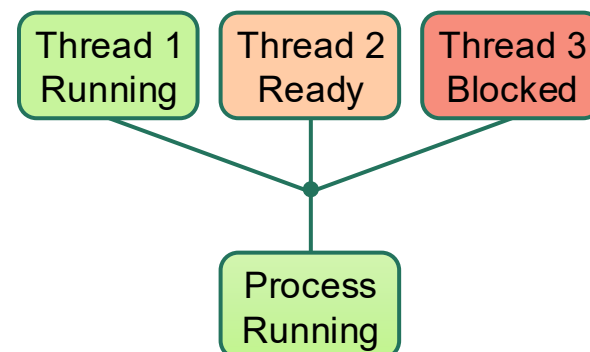
Kernel has suspended user process



Thread 3 requests I/O



Thread 3 is blocked on user-level mutex



Kernel Opacity Problem

- Kernel is completely unaware of user threads
- When user-managed thread blocks, the whole process blocks
- Kernel might take away CPU from process (e.g., time slice expired)
- User-space scheduler doesn't know these and can't schedule another thread
- One (purely academic) solution: *scheduler activations*
 - Notify user-level scheduler of relevant kernel events

Classification of OSes

- Most operating systems have either
 - One or many address spaces
 - One or many threads per address space

# threads Per AS: # of addr spaces:	One	Many
One	MS/DOS, early Macintosh	Traditional UNIX
Many	Embedded systems (Geoworks, VxWorks, JavaOS, Pilot(PC), etc.)	Mach, OS/2, Linux, Windows 10, Win NT to XP, Solaris, HP-UX, OS X

Outline

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Implementing Synchronization Objects

Programs	Bounded Buffers
Synch Objects	Mutex Semaphore Monitor
Atomic Inst	Load/Store Disable Interrupts Test&Set

Atomic Memory Operations

- In most architectures, load and store operations on single byte are **atomic**
 - Threads cannot get context switched in middle of load/store to/from a *word*
- In x86, load and store operations on **naturally aligned** variables are atomic
 - I.e., aligned to an address at least multiple of its own size
 - E.g., 8-byte int that is aligned to an address that's multiple of 8
- **Many instructions are not atomic**
 - Double-precision floating point store often not atomic
 - VAX and IBM 360 had an instruction to copy whole array
- Unless otherwise stated, we assume loads and stores are **atomic**

Mutual Exclusion With Load and Store

// Thread A

```
valueA = BUSY;
```

```
turn = 1;
```

```
while (valueB == BUSY  
      && turn == 1);
```

// critical section

```
valueA = FREE;
```

// Thread B

```
valueB = BUSY;
```

```
turn = 0;
```

```
while (valueA == BUSY  
      && turn == 0);
```

// critical section

```
valueB = FREE;
```

Mutual Exclusion

With Load and Store: Discussion

- It's very unsatisfactory
 - Only protects a **single critical section**
 - Extending this solution to protect multiple critical sections is nontrivial
 - Way too **complex** even for this simple example
 - It's hard to convince yourself that this really works
 - A's **code is different from** B's
 - What if there are more than two threads?
 - See **Peterson's algorithm**
 - A is **busy-waiting** while B is in critical section
 - While A is waiting, it is consuming CPU time
- It doesn't work on most of today's computers
 - It only works if instructions are executed in program order
 - Compilers and hardware could **reorder** instructions
 - E.g., store buffer allows next instruction to execute while store is being completed

Question: Can This Panic?

// Thread 1

```
p = someComputation();  
pInitialized = true;
```

// Thread 2

```
While (!pInitialized);  
q = someFunc(p);  
If (q != someFunc(p))  
    panic();
```

Aside: Memory Consistency in Multiprocessors

```
// initially flag = data = r1 = r2 = 0
```

CPU1

S1: data = NEW;

S2: flag = SET;

CPU2

L1: r1 = flag;

B1: if (r1 != SET) goto L1;

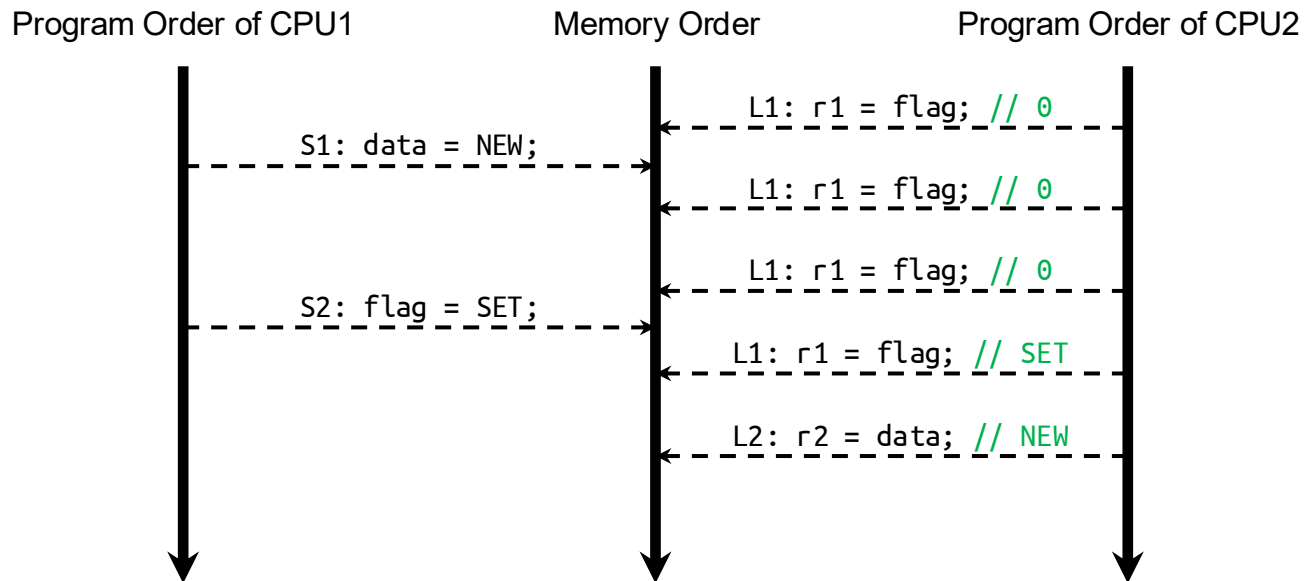
L2: r2 = data;

- Intuition says we should print $r2 = \text{NEW}$
- Yet, in some consistency models, this isn't required!

Sequential Consistency

“The result of any execution is the same as if the operations of all processors (CPUs) were executed in some sequential order, and the operations of each individual processor (CPU) appear in this sequence in the order specified by its program.”

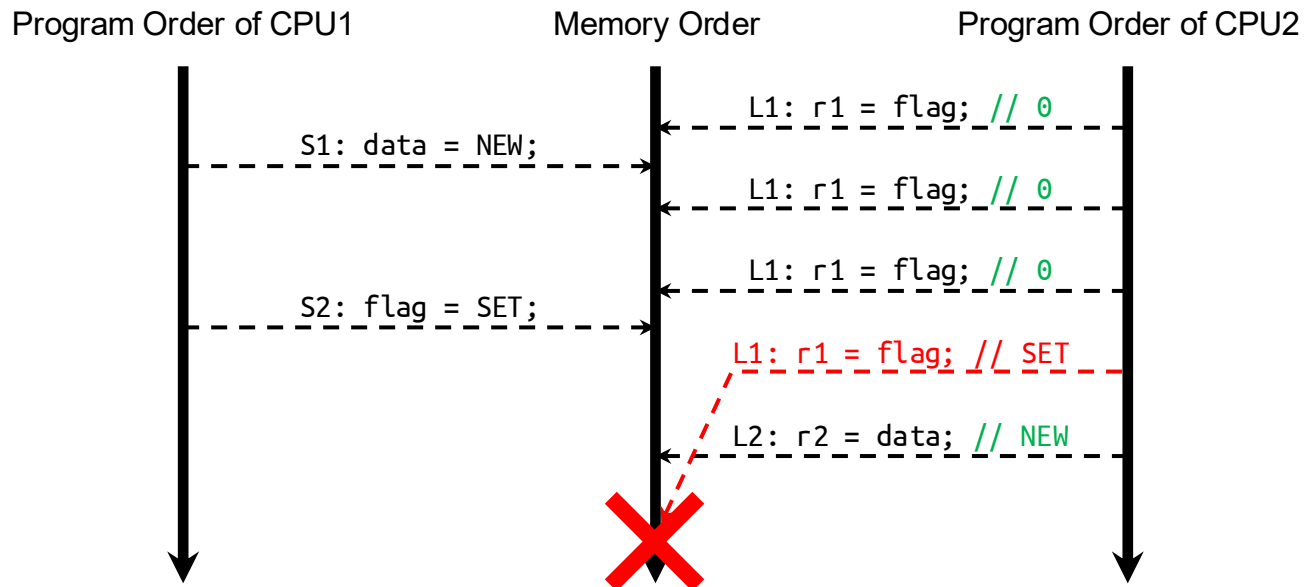
Lamport, 1979



Sequential Consistency (cont.)

“The result of any execution is the same as if the operations of all processors (CPUs) were executed in some sequential order, and the operations of each individual processor (CPU) appear in this sequence in the order specified by its program.”

Lamport, 1979



x86 Memory-consistency Model

// initially $x = y = r1 = r2 = r3 = r4 = 0$

CPU1

S1: $x = \text{NEW};$

L1: $r1 = x;$

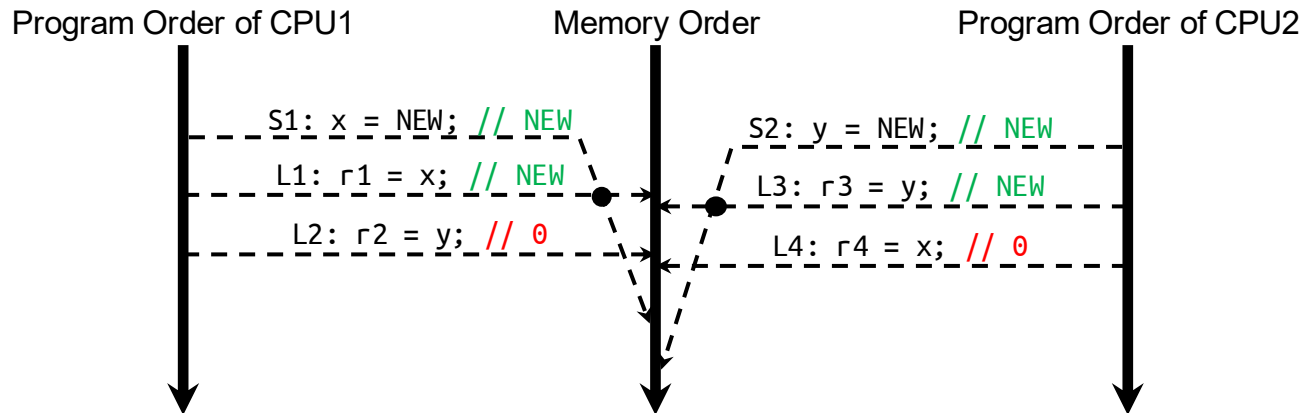
L2: $r2 = y;$

CPU2

S2: $y = \text{NEW};$

L3: $r3 = y;$

L4: $r4 = x;$



Mutual Exclusion with Mutex

- Mutex operations
 - `mutex.lock()` – wait until lock is free, then grab
 - `mutex.unlock()` – Unlock, waking up anyone waiting
- Rules of using mutex
 - Always lock before accessing shared data
 - Best place for locking is **beginning** of procedure!
 - Always unlock after finishing with shared data
 - Best place for unlocking is **end** of procedure!
 - Only the thread that locked mutex should unlock it
 - **DO NOT** throw locked mutex to someone else to unlock
 - Never access shared data without lock
 - **Danger! Don't do it even if it's tempting!**

Lock Before Accessing Shared Data, ALWAYS!

```
getP() {  
    if (p == NULL) {  
        mutex.lock();  
        if (p == NULL) {  
            temp = malloc(sizeof(...));  
            temp->field1 = ...;  
            temp->field2 = ...;  
            p = temp;  
        }  
        mutex.unlock();  
    }  
    return p;  
}
```

- Safe but expensive solution is

```
getP() {  
    mutex.lock();  
    if (p == NULL) {  
        temp = malloc(sizeof(...));  
        temp->field1 = ...;  
        temp->field2 = ...;  
        p = temp;  
    }  
    mutex.unlock();  
    return p;  
}
```

Does this work?

- No! Compiler/HW could make p point to temp before its fields are set
- This is called **double-checked locking**

Mutex Implementation - Take 1: Disabling Interrupts

- Recall: context switching is triggered in two ways
 - Voluntary: thread does something to relinquish CPU
 - Involuntary: interrupts cause dispatcher to take CPU
- On uniprocessors, we can avoid context switching by
 - Avoiding voluntary context switches
 - Preventing involuntary context switches by disabling interrupts
- Naïve implementation of mutex in uniprocessors

```
class Mutex {  
    public:  
        void lock() { disable_interrupts(); };  
        void unlock() { enable_interrupts(); };  
}
```

Problems with Naïve Implementation of Mutex

- OS cannot let users use this!

```
Mutex::lock();  
while(TRUE);
```

- It does not work well in multiprocessors
 - Other CPUs could be interrupted



- Real-time OSes should provide guarantees on timing!
 - Critical sections might be arbitrarily long
 - What happens with I/O or other important events?
 - “Reactor about to meltdown. Help?”

Implementation of Mutex - Take 2: Disabling Interrupts + Lock Variable

Key idea: maintain lock variable and impose mutual exclusion only during operations on that variable

```
class Mutex {  
    private:  
        int value = FREE;  
        Queue waiting;  
    public:  
        void lock();  
        void unlock();  
}
```

Take 2 (cont.)

```
Mutex::lock() {  
    disable_interrupts();  
    if (value == BUSY) {  
        // Add TCB to waiting queue  
        waiting.add(runningTCB);  
        runningTCB->state = WAITING;  
        // Pick new thread to run  
        chosenTCB = ready_list.get_nextTCB();  
        // Switch to new thread  
        thread_switch(runningTCB, chosenTCB);  
        // We're back! We have locked mutex!  
        runningTCB->state = RUNNING;  
    } else {  
        value = BUSY;  
    }  
    enable_interrupts();  
}
```

```
Mutex::unlock() {  
    disable_interrupts();  
    if (!waiting.empty()) {  
        // Make another TCB eady  
        next = waiting.remove();  
        next->state = READY;  
        ready_list.add(next);  
    } else {  
        value = FREE;  
    }  
    enable_interrupts();  
}
```

- Enable/disable interrupts also act as a memory barrier operation forcing all memory writes to complete first

Take 2: Discussion

- Why do we need to disable interrupts at all?
 - Avoid interruption between checking and setting lock value
 - Otherwise, two threads could think that they both have locked the

mutex

```
Mutex::lock() {  
    disable_interrupts();  
    if (value == BUSY) {  
        ...  
    } else {  
        value = BUSY;  
    }  
    enable_interrupts();  
}
```

} Critical section of mutex
(different from critical section of
program)

- Unlike previous solution, critical section (inside `lock()`) is very short
 - User of mutex can take as long as they like in **their own critical section**
(doesn't impact global machine behavior)

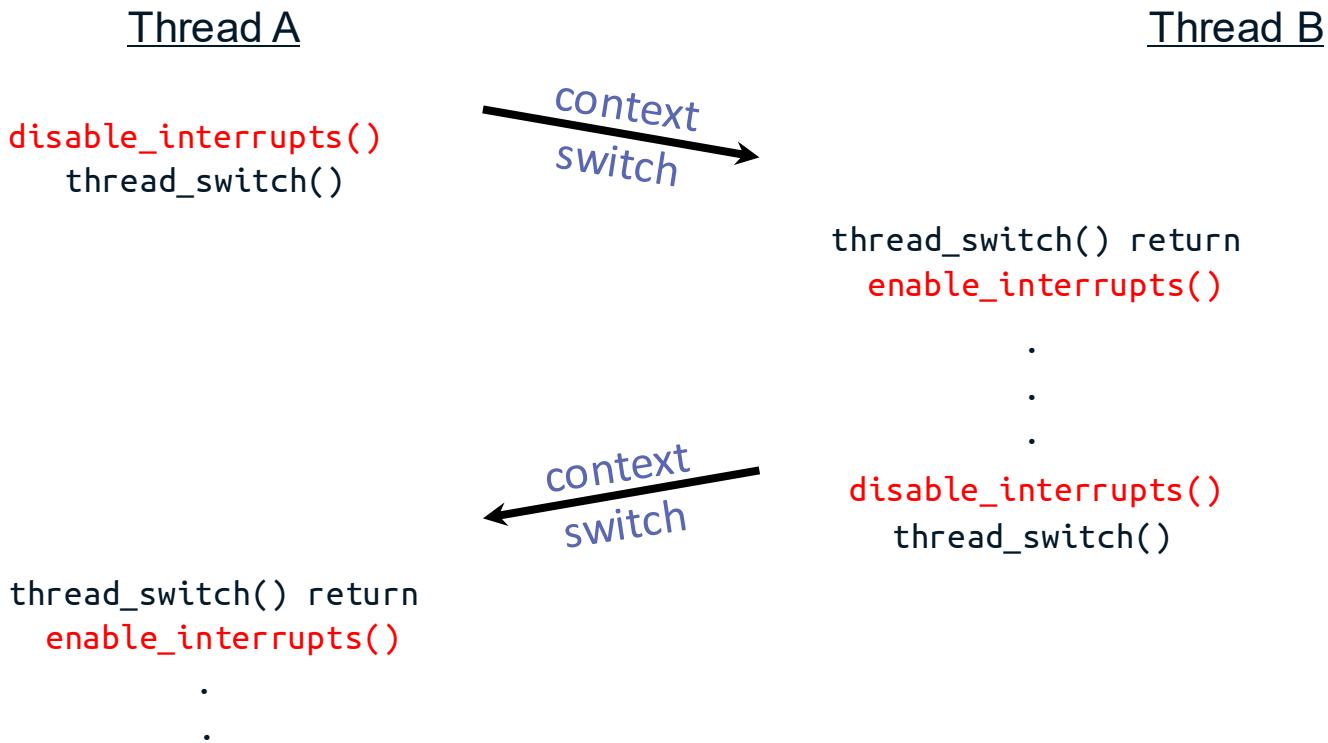
Re-enabling Interrupts

```
enable interrupts here? → Mutex::lock() {  
    disable_interrupts();  
    if (value == BUSY) {  
        waiting.add(runningTCB);  
        runningTCB->state = WAITING;  
        chosenTCB = ready_list.get_nextTCB();  
        thread_switch(runningTCB, chosenTCB);  
        runningTCB->state = RUNNING;  
    } else {  
        value = BUSY;  
    }  
    enable_interrupts();  
}
```

- Before putting thread on wait queue?
 - `unlock()` can check waiting queue and not wake up thread
- After putting thread on wait queue?
 - `unlock()` puts thread on ready queue, but thread still thinks it needs to go to sleep!
 - Thread goes to sleep while keeping mutex locked (deadlock!)
- After `thread_switch()`? But ... how?

How to Re-enable After thread_switch()?

- It is responsibility of next thread to re-enable interrupts
 - This invariant should be carefully maintained
- When sleeping thread wakes up, returns to lock() and re-enables interrupts



Problems with Take 2

- User libraries cannot use this implementation (why?)
- Doesn't work well on multiprocessor
 - Disabling interrupts on all processors requires messages and would be very time consuming
- Alternative solution: **atomic read-modify-write instructions**
 - Read value from an address and then write new value to it *atomically*
 - Make HW responsible for implementing this correctly
 - Uniprocessors (not too hard)
 - Multiprocessors (requires help from cache coherence protocol)
 - Unlike disabling interrupts, this can be used in both uniprocessors and multiprocessors

Examples of Read-Modify-Write Instructions

- ```
test&set (&address) {
 result = M[address];
 M[address] = 1;
 return result;
}
```

```
/* most architectures */
/* return result from
 "address" and set value at
 "address" to 1 */
```
- ```
swap (&address, register) {  
    temp = M[address];  
    M[address] = register;  
    register = temp;  
}
```

```
/* x86 */  
/* swap register's value to  
   value at "address" */
```
- ```
compare&swap (&address, reg1, reg2) {
 if (reg1 == M[address]) {
 M[address] = reg2;
 return success;
 } else {
 return failure;
 }
}
```

```
/* 68000 */
```

# Spinlock with test&set()

---

- Simple implementation

```
class Spinlock {
 private:
 int value = 0
 public:
 void lock() { while(test&set(value)); };
 void unlock() { value = 0; };
}
```

- Unlocked mutex: test&set reads 0 and sets value = 1
- Locked mutex: test&set reads 1 and sets value = 1 (no change)
- What is wrong with this implementation?
  - Waiting threads consume cycles while **busy-waiting**

# Spinlock with test&set(): Discussion

---

- Upside?
  - Machine can receive interrupts
  - User code can use this mutex
  - Works on multiprocessors
- Downside?
  - This is very wasteful as threads consume CPU cycles (busy-waiting)
  - Waiting threads may delay the thread that has locked mutex (no one wins!)
  - **Priority inversion**: if busy-waiting thread has higher priority than the thread that has locked mutex then there will be no progress! (more on this later)
- In semaphores and monitors, threads may wait for arbitrary long time!
  - Even if busy-waiting was OK for mutexes, it's not OK for other



# Implementation of Mutex - Take 3: Using Spinlock

---

- Can we implement mutex with test&set without busy-waiting?
  - We cannot eliminate busy-waiting, but we can minimize it!
  - **Idea:** only busy-wait to atomically check mutex value

```
class Mutex {
 private:
 int value = FREE;
 Spinlock mutex_spinlock;
 Queue waiting;
 public:
 void lock();
 void unlock();
}
```

```
class Scheduler {
 private:
 Queue readyList;
 Spinlock scheduler_spinlock;
 public:
 void suspend(Spinlock *spinlock);
 void make_ready(TCB *tcb);
}
```



# Implementation of Mutex - Take 3 (cont.)

---

```
Mutex::lock() {
 mutex_spinlock.lock();
 if (value == BUSY) {
 // Add TCB to waiting queue
 waiting.add(runningTCB);
 scheduler->suspend(&mutex_spinlock)
 // Scheduler unlocks mutex_spinlock
 } else {
 value = BUSY;
 }
 mutex_spinlock.unlock();
}
```

```
Mutex::unlock() {
 mutex_spinlock.lock();
 if (!waiting.empty()) {
 // Make another TCB ready
 next = waiting.remove();
 scheduler->make_ready(next);
 } else {
 value = FREE;
 }
 mutex_spinlock.unlock();
}
```

Can interrupt handler use this lock?

- No! Interrupt handler is not a thread, it cannot be suspended

How should we protect data shared by interrupt handler and kernel thread?

- Use spinlocks!
- To avoid deadlock, kernel thread should disable interrupts before locking the spinlock.
- Otherwise, interrupt handler could spin forever if spinlock is locked by a kernel

# Implementation of Mutex - Take 3 (cont.)

---

```
Scheduler::suspend(Spinlock *spinlock) {
 disable_interrupts();
 scheduler_spinlock.lock();
 spinlock->unlock();
 runningTCB->state = WAITING;
 chosenTCB = ready_list.get_nextTCB();
 thread_switch(runningTCB, chosenTCB);
 runningTCB->state = RUNNING;
 scheduler_spinlock.unlock();
 enable_interrupts();
}
```

```
Scheduler::make_ready(TCB *tcb) {
 disable_interrupts();
 scheduler_spinlock.lock();
 ready_list.add(tcb);
 thread->state = READY;
 scheduler_spinlock.unlock();
 enable_interrupts();
}
```

Why disable interrupts?

- To avoid **deadlock**!
- Interrupt handler could spin forever if it needs scheduler's spinlock!

What might happen if we unlock `mutex_spinlock` before `suspend()`?



# Mutex Using Interrupts vs. Spinlock

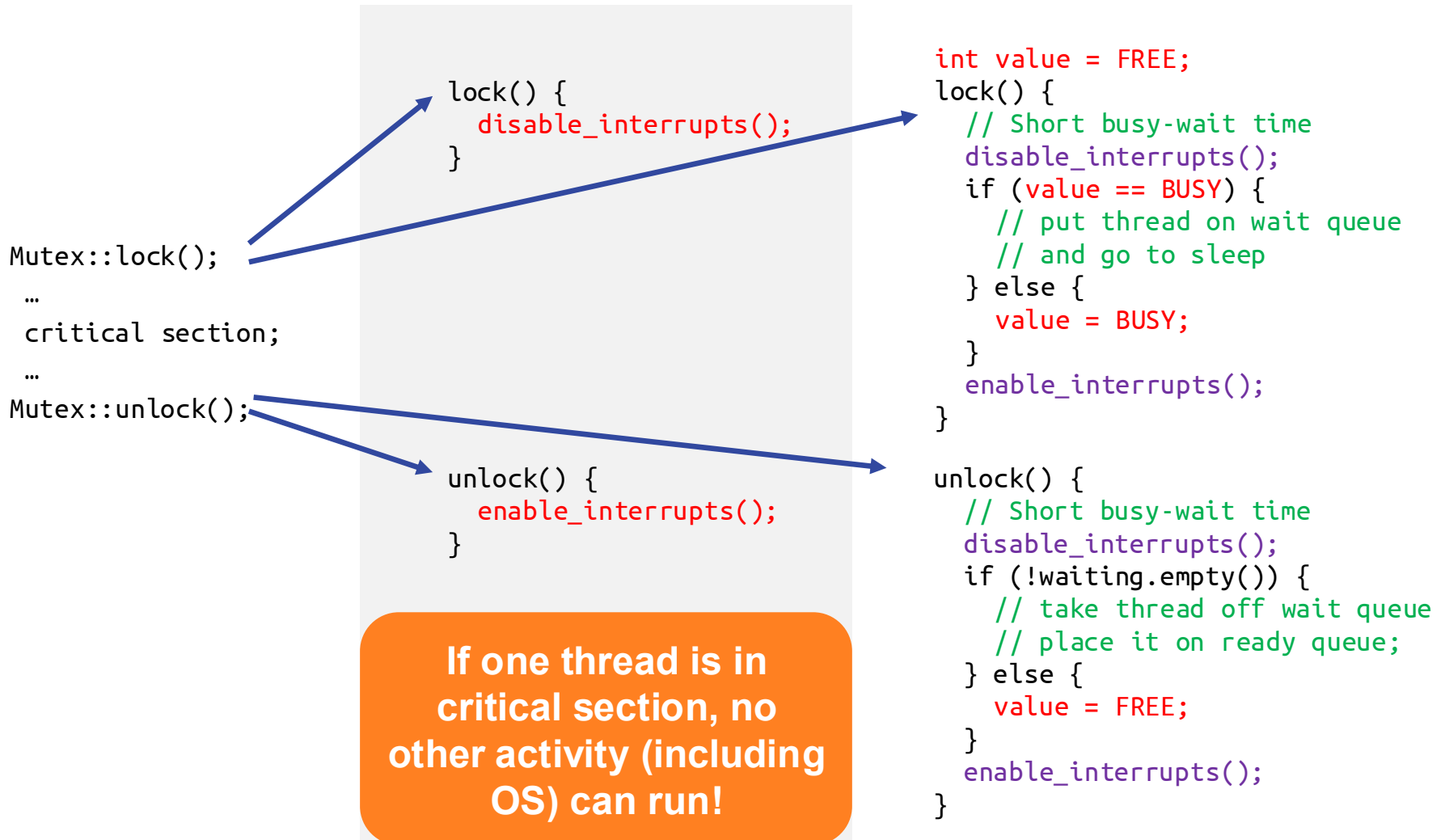
---

```
lock() {
 disable_interrupts();
 if (value == BUSY) {
 // put thread on wait queue and
 // go to sleep
 } else {
 value = BUSY;
 }
 enable_interrupts();
}
```

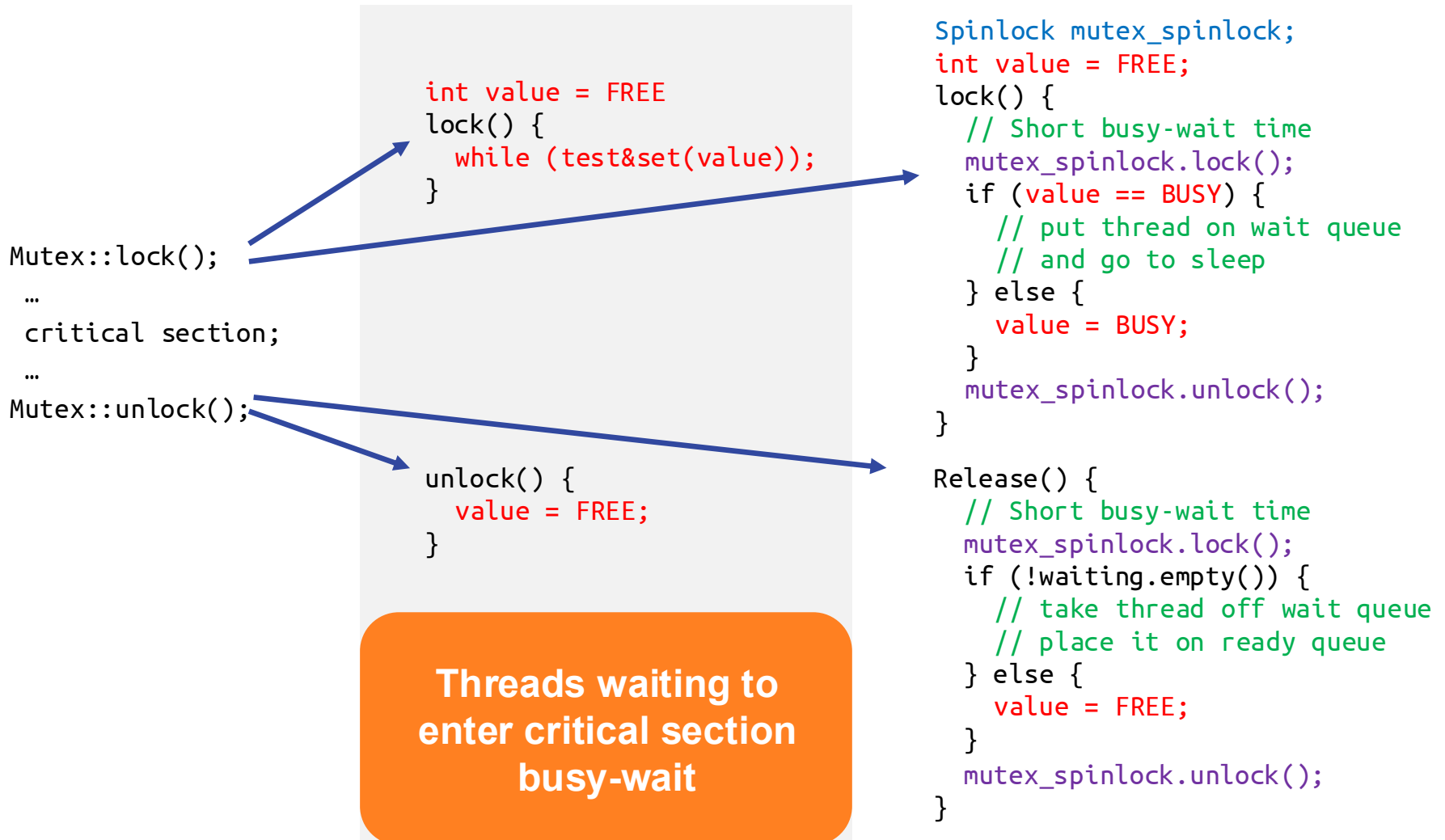
```
lock() {
 mutex_spinlock.lock();
 if (value == BUSY) {
 // put thread on wait queue and
 // go to sleep
 } else {
 value = BUSY;
 }
 mutex_spinlock.unlock();
}
```

- Replace
  - disable interrupts;  $\Rightarrow$  spinlock.lock;
  - enable interrupts  $\Rightarrow$  spinlock.unlock;

# Recap: Mutexes Using Interrupts



# Recap: Mutexes Using Spinlock (test&set)



# Mutex Implementation in Linux

---

- Most mutexes are free most of the time
  - Linux implementation takes advantage of this fact
- Hardware supports powerful atomic operations
  - E.g., atomic increment, decrement, exchange, etc.
  - Linux implementation takes advantage of these too
- Fast path
  - If mutex is unlocked, and no one is waiting, two instructions to lock
  - If no one is waiting, two instructions to unlock
- Slow path
  - If mutex is locked or someone is waiting, use take 3 implementation

# Mutex Implementation in Linux (cont.)

---

```
struct Mutex {
 // 1: unlocked; < 1: locked
 atomic_t count;
 Spinlock mutex_spinlock;
 Queue waiting;
}

// code for lock()
lock decl (%eax)
// jump if not signed
// i.e., if value is now 0
jns 1f
call slow_path_lock
1:
//critical section
```

- For `Mutex::lock()`, Linux uses *macro*
  - To void making procedure call on fast path
- x86 *lock* prefix before *decl* instruction signifies to processor that instruction should be executed atomically

# Mutex Implementations: Discussion

---

- Our lock implementations are procedure calls
- Work well for kernel threads
- Does not work properly for user threads
  - Because system call may often disable interrupts/save state to TCB
  - But same basic idea works – e.g., in Linux, user-level mutex has two paths - Fast path: lock using test&set and slow path: system call to kernel, use kernel mutex
- How do lock-initiated and timer-interrupt-initiated switches interleave?
  - Turns out, they just work as long as we maintain the invariant on interrupts - disable before calling `thread_switch()` and enable when `thread_switch()` returns

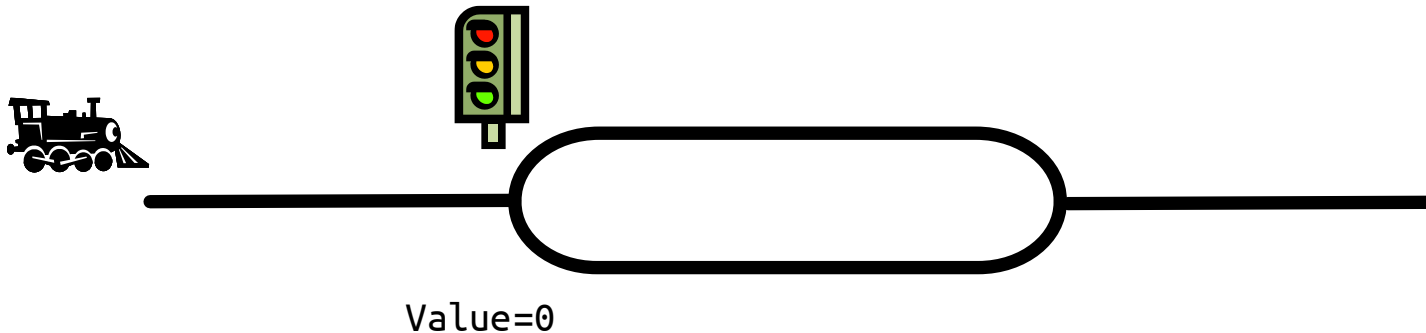




# Recall: Semaphores

---

- First defined by Dijkstra in late 60s
- Main synchronization primitive used in original UNIX
- Semaphore has non-negative integer value and 2 operations
  - **P()**: atomic operation that waits for semaphore to become positive, then decrements it by one
  - **V()**: atomic operation that increments semaphore by one, then wakes up a waiting P(), if any



# Implementation of Semaphore

---

```
Semaphore::P() {
 semaphore_spinlock.lock();
 if (value == 0) {
 waiting.add(myTCB);
 scheduler->suspend(&semaphore_spinlock);
 } else {
 value--;
 }
 semaphore_spinlock.unlock();
}
```

```
Semaphore::V() {
 semaphore_spinlock.lock();
 if (!waiting.empty()) {
 next = waiting.remove();
 scheduler->make_ready(next);
 } else {
 value++;
 }
 semaphore_spinlock.unlock();
}
```

Can interrupt handler use this semaphore?

- It cannot use P (why?), but it might want to use V (more on this later)
- In that case, interrupts should be disabled at the beginning of P and V and enabled at the end

# Semaphores are Harmful!

---

“During system conception it transpired that we used the semaphores in **two completely different ways**. The difference is so marked that, looking back, one wonders whether it was really fair to present the two ways as uses of the very same primitives. On the one hand, we have the semaphores used for mutual exclusion, on the other hand, the private semaphores.”

Dijkstra “The structure of the 'THE'-Multiprogramming System” *Communications of the ACM* v. 11 n. 5 May 1968.

# Monitors and Condition Variables

---

- **Problem:** analyzing code that uses semaphores is complex
  - They are dual purpose (both mutual exclusion and scheduling constraints)
- **Solution:** use monitors
  - It consists of one mutex with zero or more **condition variables (CV)**
  - Mutex is used for mutual exclusion, CV's are used for scheduling constraints

# Recall: Condition Variables

---

- CV is queue of threads waiting for an event inside critical section
  - Makes it possible to go to sleep inside critical section
  - Atomically unlocks mutex at time thread goes to sleep
  - With semaphores, threads cannot wait inside critical section (deadlock)
- CV operations
  - `wait(Mutex *CVMutex)`
    - Atomically unlocks mutex, puts thread to sleep, and relinquishes processor
    - Attempts to locks mutex when thread wakes up
  - `signal()`
    - Wakes up a waiter, if any
  - `broadcast()`
    - Wakes up all waiters, if any

# Recall: Properties of CV

---

- Condition variables are **memoryless**
  - No internal memory except a queue of waiting threads
  - No effect in calling `signal/broadcast` on empty queue
- **ALWAYS** hold lock when calling `wait()`, `signal()`, `broadcast()`
- Calling `wait()` **atomically** adds thread to wait queue and releases lock
- Re-enabled waiting threads may not run immediately
  - No atomicity between `signal/broadcast` and the return from `wait`



# Recall: CV Design Pattern

---

```
method_that_waits() {
 mutex.lock();

 // Read/write shared state

 while (!testSharedState())
 cv.wait(&mutex);

 // Read/write shared state

 mutex.unlock();
}
```

```
method_that_signals() {
 mutex.lock();

 // Read/write shared state

 // If testSharedState is now true
 cv.signal();

 // Read/write shared state

 mutex.unlock();
}
```

# Example: Bounded Buffer With Monitors

---

```
Mutex BBMutex;
CV emptyCV, fullCV;
```

```
produce(item) {
 BBMutex.lock();
 while (buffer.size() == MAX)
 fullCV.wait(&BBMutex);
 buffer.enqueue(item);
 emptyCV.signal();
 BBMutex.unlock();
}
```

```
consume() {
 BBMutex.lock();
 while (buffer.isEmpty())
 emptyCV.wait(&BBMutex);
 item = buffer.dequeue();
 fullCV.signal();
 BBMutex.unlock();
 return item;
}
```

// lock the mutex  
// wait until there is space

// signal waiting costumer  
// unlock the mutex

// lock the mutex  
// wait until there is item

// signal waiting producer  
// unlock the mutex



# Mesa vs. Hoare Monitors

---

- Consider piece of `consume()` code

```
while (queue.empty())
 emptyCV.wait(&mutex);
```

- Why didn't we do this?

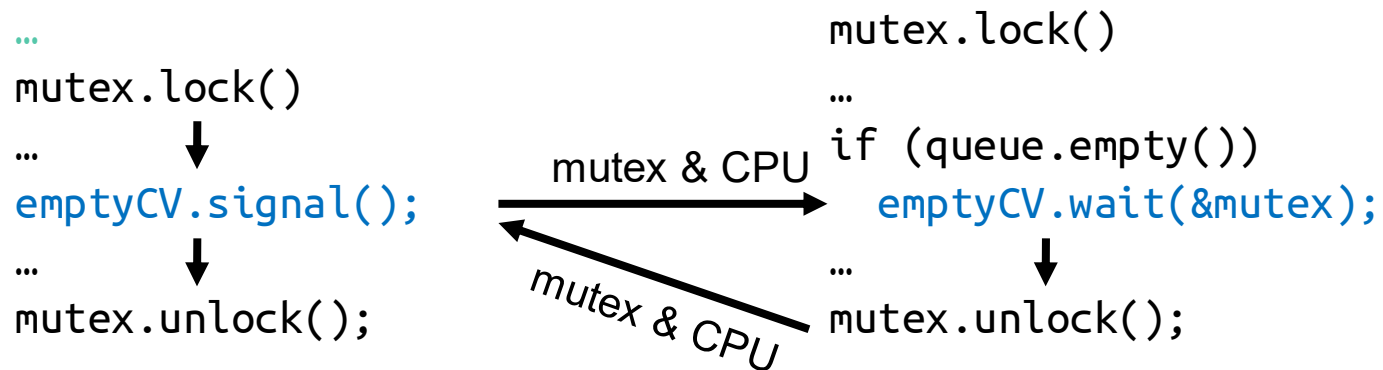
```
if (queue.empty())
 emptyCV.wait(&mutex);
```

- **Answer:** it depends on the type of scheduling
  - Hoare style
  - Mesa style

# Hoare Monitors

---

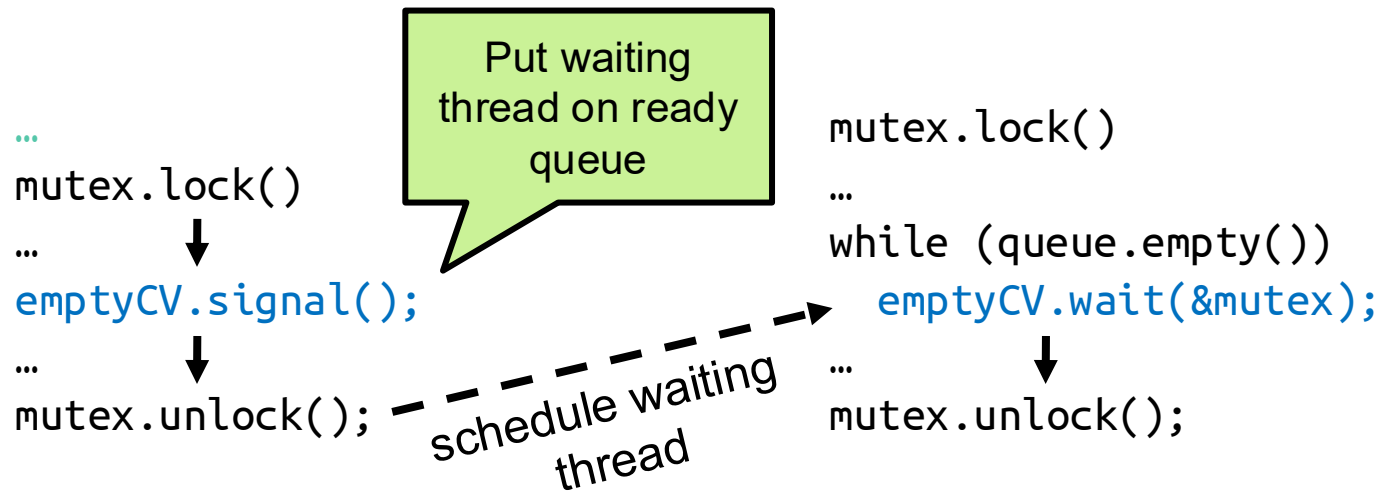
- Signaler gives up mutex and processor to waiter – waiter runs immediately
- Waiter gives up mutex and processor back to signaler when it exits critical section or if it waits again



# Mesa Monitors

---

- Signaler keeps mutex and processor
- Waiter placed on ready queue with no special priority
- Practically, need to check condition again after wait
- Most real operating systems



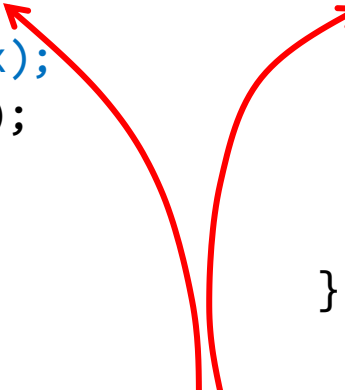
# Mesa Monitor: Why “while()”?

---

- What if we use “if” instead of “while” in bounded buffer example?

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

```
produce(item) {
 mutex.lock();
 if (queue.size() == MAX)
 fullCV.wait(&mutex);
 queue.add(item);
 emptyCV.signal();
 mutex.unlock();
}
```



Use “if” instead of “while”

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: unlocked  
emptyCV queue →  
NULL

CPU State

Running: **T1**  
ready queue → NULL  
...

T1 (**Running**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: **locked (T1)**  
emptyCV queue →  
NULL

CPU State

Running: T1  
ready queue → NULL  
...

T1 (Running)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

App. Shared State

queue



Monitor

mutex: **unlocked**  
emptyCV queue → **T1**

CPU State

Running:  
ready queue → NULL  
...

T1 (**Waiting**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

wait(&lock) puts  
thread on emptyCV  
queue and releases  
lock

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: unlocked  
emptyCV queue → T1

CPU State

Running: T2  
ready queue → NULL  
...

T1 (Waiting)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T2 (Running)

```
produce(item) {
 mutex.lock();
 if (queue.size() == MAX)
 fullCV.wait(&mutex);
 queue.add(item);
 emptyCV.signal();
 mutex.unlock();
}
```



# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: **locked (T2)**  
emptyCV queue → T1

CPU State

Running: T2  
ready queue → NULL  
...

T1 (Waiting)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T2 (Running)

```
produce(item) {
 mutex.lock();
 if (queue.size() == MAX)
 fullCV.wait(&mutex);
 queue.add(item);
 emptyCV.signal();
 mutex.unlock();
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

App. Shared State

queue



Monitor

mutex: locked (T2)  
emptyCV queue →  
**NULL**

CPU State

Running: T2  
ready queue → **T1**  
...

T1 (**Ready**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T2 (Running)

```
produce(item) {
 mutex.lock();
 if (queue.size() == MAX)
 fullCV.wait(&mutex);
 queue.add(item);
 emptyCV.signal();
 mutex.unlock();
}
```

signal() wakes up  
and moves it to ready  
queue

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: locked (T2)  
emptyCV queue →  
NULL

CPU State

Running: T2  
ready queue → T1, T3  
...

T1 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T2 (Running)

```
produce(item) {
 mutex.lock();
 if (queue.size() == MAX)
 fullCV.wait(&mutex);
 queue.add(item);
 emptyCV.signal();
 mutex.unlock();
}
```

T3 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: **unlocked**  
emptyCV queue →  
NULL

CPU State

Running:  
ready queue → T1, T3  
...

T1 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T2 (**Terminated**)

```
produce(item) {
 mutex.lock();
 if (queue.size() == MAX)
 fullCV.wait(&mutex);
 queue.add(item);
 emptyCV.signal();
 mutex.unlock();
}
```

T3 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

App. Shared State

queue



Monitor

mutex: unlocked  
emptyCV queue →  
NULL

CPU State

Running: **T3**  
ready queue → T1

T1 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T3 is scheduled first

T3 (**Running**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

App. Shared State

queue



Monitor

mutex: **locked (T3)**  
emptyCV queue →  
NULL

CPU State

Running: T3  
ready queue → T1  
...

T1 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T3 (Running)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: locked (T3)  
emptyCV queue →  
NULL

CPU State

Running: T3  
ready queue → T1  
...

T1 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T3 (Running)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

App. Shared State

queue



Monitor

mutex: **unlocked**  
emptyCV queue →  
NULL

CPU State

Running:  
ready queue → T1  
...

T1 (Ready)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

T3 (**Terminated**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```



# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: **locked (T1)**  
emptyCV queue →  
NULL

CPU State

Running: **T1**  
ready queue → **NULL**  
...

T1 (**Running**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: locked (T1)  
emptyCV queue →  
NULL

CPU State

Running: T1  
ready queue → NULL  
...

T1 (Running)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

Error!

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue



Monitor

mutex: locked (T1)  
emptyCV queue →  
NULL

CPU State

Running: T1  
ready queue → NULL  
...

T1 (Running)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

Check again  
if empty!

# Mesa Monitor: Why “while()”?

## (cont.)

---

App. Shared State

queue

Monitor

mutex: **unlocked**  
emptyCV queue → **T1**

CPU State

Running:  
ready queue → NULL  
...

T1 (**Waiting**)

```
consume() {
 mutex.lock();
 if (queue.empty())
 emptyCV.wait(&mutex);
 item = queue.remove();
 fullCV.signal();
 mutex.unlock();
 return item;
}
```

# Mesa Monitor: Why “while()”?

## (cont.)

---

When waiting upon a *Condition*, a **spurious wakeup** is permitted to occur, in general, as a concession to the underlying platform semantics. This has little practical impact on most application programs as a *Condition* should always be waited upon in a loop, testing the state predicate that is being waited for

From Java User Manual

# Condition Variable vs. Semaphore

---

- CV's `signal()` has **no memory**
  - If `signal()` is called before `wait()`, then signal is waisted
- Semaphore's `v()` has memory
  - If `v()` is called before `P()`, `P()` will not wait
- Generally, it's better to use monitors but not always
- Example: interrupt handlers
  - Shared memory is used concurrently by interrupt handler and kernel thread
  - Interrupt handler cannot use mutexes
  - Kernel thread checks for data and calls `wait()` if there is no data
  - Interrupt handler write to shared memory and calls `signal()`
    - This is called **naked notify** because interrupt handler hasn't locked mutex (why?)
  - This may not work if signal comes before kernel thread calls `wait`
  - Common solution is to use semaphores instead

# Implementation of Condition Variables

---

```
class CV {
 private:
 Queue waiting;
 public:
 void wait(Mutex *mutex);
 void signal();
 void broadcast();
}

CV::wait(Mutex *mutex) {
 waiting.add(myTCB);
 scheduler.suspend(&mutex);
 mutex->lock();
}

CV::signal() {
 if (!waiting.empty()) {
 thread = waiting.remove();
 scheduler.make_ready(thread);
 }
}

void CV::broadcast() {
 while (!waiting.empty()) {
 thread = waiting.remove();
 scheduler.make_ready(thread);
 }
}
```

Why doesn't class `CV` need `cv_spinlock`?

- Since `mutex` is locked whenever `wait`, `signal`, or `broadcast` is called, we already have mutually exclusive access to condition wait queue

# Implementation of Condition Variable using Semaphores (Take 1)

---

```
wait(*mutex) {
 mutex->unlock();
 semaphore.P();
 mutex->lock();
}

signal() {
 semaphore.V();
}
```

- Does this work?
  - No! signal() should not have memory!



# Implementation of Condition Variable using Semaphores (Take 2)

---

```
wait(*mutex) {
 mutex->unlock();
 semaphore.P();
 mutex->lock();
}

signal() {
 if (semaphore's queue is not empty)
 semaphore.V();
}
```

- Does this work?
  - No! For one, not legal to look at contents of semaphore's queue.
  - But also, unlocking mutex and going to sleep should happen atomically – signaler can slip in after mutex is unlocked, and before waiter is put on wait queue, which means waiter never

# Implementation Condition Variable using Semaphores (Take 3)

---

Key idea: have separate semaphore for each waiting thread

and put semaphores in ordered queue

```
wait(*mutex) {
 semaphore = new Semaphore; // a semaphore per waiting thread
 queue.add(semaphore); // queue for waiting threads
 mutex->unlock();
 semaphore.P();
 mutex->lock();
}

signal() {
 if (!queue.empty()) {
 semaphore = queue.remove()
 semaphore.V();
 }
}
```

# Summary

---

- Use HW atomic primitives as needed to implement synchronization
  - Disabling of Interrupts, test&set, swap, compare&swap
- Define lock variable to implement mutex,
  - Use HW atomic primitives to protect modifications of that variable
- Maintain the invariant on interrupts
  - Disable interrupts before calling `thread_switch()` and enable them when `thread_switch()` returns
- Be very careful not to waste machine resources
  - Shouldn't disable interrupts for long
  - Shouldn't busy-wait for long

# Questions?

---



# Acknowledgment

---

- Slides by courtesy of Anderson, Culler, Stoica, Silberschatz, Joseph, and Canny